MILITARY SPECIFICATION

PADDINGS AND COVERS FOR FLATWORK IRONER ROLL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the requirements for fiber and metal mesh padding and fiber covering material used as replacement covering set for flatwork ironers (see 6.1).

1.2 Classification.- Paddings shall be of the following types and classes as specified (see 6.2).

Type I - Fiber felted, knitted or batting pad.

Type II - Metal mesh padding.

Class 1 - 110 inches long.
Class 2 - 120 inches long.
Class 3 - 126 inches long.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-45 - Tape, Gummed, Paper, Reinforced And Plain, For Sealing And Securing.

FSC 3510

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MILITARY

MIL-L-10547 - Liners, Case And Sheet, Overwrap; Water Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Method.
FED-STD-751 - Stitches, Seams, and Stitchings.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Standards

ASTM D-1061-68 - Standard Specification for Asbestos Lap.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Padding material.-

3.1.1 Fiber padding.- The padding shall be a felted, knitted or batting pad made of not less than 60 percent asbestos when tested as specified in 4.4.1. The balance of the fiber content can be cotton, synthetic or a mixture of cotton and synthetic fiber. The padding shall be of sufficient width for one wrap around the roll. The padding shall be of such thickness that when the binder, padding and cover are installed on the roll, the diameter of the entire assembly shall be within the limits recommended by the manufacturer of the flatwork ironer.

3.1.2 Metal mesh padding.- The metal mesh padding furnished under this specification shall be the manufacturer's current standard product except for any changes required to comply with the requirements of this specification. All accessories and components shall be furnished with paddings that are normally furnished with the standard product offered commercially.

3.1.2.1 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for padding provided the completed item complies with all provisions of this specification.

3.1.2.2 Corrosion-resisting steel.- Corrosion-resisting steel shall be any steel number in the 300 or 400 series designation.

3.1.2.3 Design and construction.- Metal mesh padding for flatwork ironer rolls shall be fabricated of steel specified in 3.1.2.2, and shall be a once around pad. The pad shall be tapered on the ends to conform to the ironing area used on the flatwork ironer rolls.

3.1.2.4 The finished fabricated metal mesh padding shall be free of sharp edges, burrs or slivers.

3.2 Binder and cover.-

3.2.1 Binder.- The binder shall be made from an uncoated woven asbestos cloth containing not less than 75 percent asbestos when tested as specified in 4.4.1. The balance of the fiber shall be cotton or a mixture of cotton and a synthetic fiber. The cloth shall have a minimum breaking strength of 71 pounds in the warp and 123 pounds in the filling when tested as specified in 4.4.1.

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3.2.2 Cover.- The cover shall be fabricated from woven asbestos cloth and shall meet the performance characteristics specified below when tested as specified in 4.4.1. The cloth, prior to processing, shall contain not less than 80 percent asbestos. The balance of the fiber content shall be cotton, a mixture of cotton and a synthetic fiber, or a synthetic fiber. The cover shall be coated or impregnated with a thermosetting resin and shall not stain or otherwise mar the appearance of the articles processed through the ironer. The cover shall have a minimum breaking strength of 190 pounds in the warp and 152 pounds in the filling and shall retain a minimum of 50.0 percent of its original breaking strength after heat ageing.

3.2.3 Cover and binder.- One end of the binder shall be attached to metal mesh padding by a mechanical fastener (clipper lace), a "T" or bar pin, and the other end to the cover by stitching. The overall wrap-around dimension of the cover and binder when attached shall have a combined or continuous length suitable for the roll it is intended to cover.

3.3 Liner.- For rolls equipped with springs, a liner conforming to the following requirements shall be supplied. The liner shall be made from a woven asbestos cloth containing not less than 75 percent asbestos when tested as specified in 4.4.1. The balance of the fiber shall be cotton, a mixture of cotton and a synthetic, or a synthetic fiber. The width of the cloth shall be sufficient to wrap once around the roll.

3.4 Thread.- The thread shall be made of a 60 percent minimum asbestos content fiber composition, or, as an alternate, a nylon, spun staple, non-melting fiber.

3.4.1 Thread asbestos.- The asbestos thread shall contain a minimum of 60 percent asbestos in combination with either cotton or a synthetic fiber or both and shall be not finer than 620 yards per pound with a minimum single strand breaking strength of 12.5 pounds when tested as specified in 4.3.1.

3.4.2 Thread, nylon, spun staple, non-melting.- The nylon spun staple, non-melting thread shall be ticket number 12, 4 ply, natural color, soft finish, z-twist and shall be not finer than 2,800 yards per pound with a minimum single strand breaking strength of 9.5 pounds when tested as specified in 4.3.1.

3.5 Fitting.- The padding and covering set required by this specification shall fit the makes of flatwork ironers specified (see 6.2) for the respective number and width of the rolls, type and diameter of rolls (spring pad or solid).

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3.6 Stitching.- The binder and the cover shall be joined using the sewing thread specified in 3.4. Stitching shall conform to stitch type 301 of FED-STD-751, 3 to 5 stitches per inch. Thread breaks in stitching shall be secured by stitching back of the break not less than 1 inch. Thread tension shall be maintained so that there will be no loose or tight stitching. All thread ends shall be trimmed.

3.7 Assembly.- The assembly of the padding, to the other components, shall be complete without broken or incomplete stitching and broken or loosely attached mechanical fastening. The assembly when attached to ironer rolls shall fit snugly without buckling or distortion.

3.8 Workmanship.- The finished padding and cover set shall conform to the quality and grade of product established by this specification. The padding and cover set shall be free of defects listed herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3 Visual examination.-

4.3.1 Visual examination of padding.-

4.3.1.1 Visual examination of fiber padding.- Each pad shall be examined. Any pad containing one or more of the following defects shall be replaced.

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Felted - Cut, hole, tear, thin place or lump.
Knitted - Cut, hole, run or dropped stitches.

4.3.1.2 Visual examination of metal mesh padding.- The finished padding shall be examined for defects in construction and workmanship. Defects shall be as classified in table I. The sample unit shall be one finished pad. The lot size shall be expressed in units of one pad each. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 defects expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect
Construction and workmanship	Ends not tapered.
	Padding split.
	Not free of punctures or tears.
	Not free of sharp edges, burrs or slivers.

4.3.2 Visual examination of binder, cover and liner.- Each binder, cover and liner shall be examined. Any of the following listed items containing one or more of the following defects shall be replaced.

Liner and Binder - Hole, cut, tear, break, smash, thick place.

Cover - Hole, cut, tear, abrasion, crease, or wrinkle, lump, foreign matter; unevenly coated or uncoated area, stain, discolored or soiled.

General - Width of any component less than the circumference of the roll.

4.3.3 Visual examination of stitched cover and binder.- Each stitched cover and binder shall be examined. Any cover and binder containing one or more of the following defects shall be replaced.

Stitching - Not specified stitch type.
- Stitches per inch more or less than specified.
- Thread break stitched less than 1 inch beyond ends of break.
- Ends of stitching backtacked less than 1/2 inch.
- Loose tension resulting in a loose seam.
- Tight tension (stitches break when normal strain is applied to the seam or stitching).
- Thread ends not trimmed throughout.

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4.3.4 Visual examination of metal mesh padding, cover and binder assembly.- The completely assembled end item shall be examined for defects in construction and workmanship. Defects shall be as classified in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Defect	Classification Major	Minor
Stitching broken.	X	
Stitching incomplete.		X
Mechanical fastening broken.	X	
Mechanical fastening loosely attached.		X
Assembly not free of buckling or distortion when attached to ironer roll.	X	
Binder and cover not as specified.	X	

4.3.5 Dimensional examination.- Examination shall be made of the end item to determine compliance with dimensional requirements. Any nonconformance with specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects expressed in terms of defects per hundred units.

4.3.6 Examination of preparation for delivery requirements.- An examination shall be made to determine that the packaging, packing and marking complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as: Incomplete closure of container flaps, loose straping, improper taping, inadequate stapling or reinforcing. Bulged or distorted container.
Weight	Weight of contents is more than specified.

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4.4 Component testing.- The components listed in 4.4.1 shall be tested for the characteristics listed. Testing shall be accomplished on the materials for the components prior to fabricating into the specified components of the end item.

4.4.1 Test.- The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 1/2 yard full width of the padding, liner and cover cloth, and 1 yard full width of the binder cloth. The sample unit for thread shall be one cone, tube or spool. The sample size (number of sample units) shall be as shown in table III. The lot shall be expressed in units of 1 yard for the padding, liner, cover cloth and binder cloth and the thread shall be expressed in units of one cone, tube or spool. The lot shall be unacceptable if one or more units fail to meet any requirements specified.

TABLE III

Lot size (units)	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE IV

Components	Characteristics	Test method
Padding	Material identification	1/
	Asbestos fiber content	4.4.1.1
Binder	Material identification	1/
	Asbestos fiber content	4.4.1.1
	Breaking strength (warp and filling)	5100
Liner	Material identification	1/
	Asbestos fiber content	4.4.1.1

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TABLE IV (cont'd)

Components	Characteristics	Test method
Cover		
Before processing	Material identification	1/
	Asbestos fiber content	4.4.1.1
After processing	Staining	4.4.1.2
	Breaking strength (warp and filling)	
	Original	5100
	After heat ageing	5850 2/
	Coating identification	1/
Thread, all	Material identification	1/
	Breaking strength	4100
	Length per pound	4010
Thread 60 percent asbestos	Asbestos content	4.4.1.1
Thread, nylon spun staple, non-melting	Color	3.4 1/
	Finish	3.1.5.2 1/
	Twist direction	4050.1
	Ticket number	3.4 1/
	Number of plies	Visual 2/

1/ Acceptance shall be based on supplier's certification of compliance. The certificate shall be accompanied by actual test, inspection, or other verifiable quality data.

2/ One determination and the result reported as "pass or fail". Except that the specimen shall be heated for 24 hours at an oven temperature of 400°F.

4.4.1.1 Asbestos fiber content.- Asbestos fiber content shall be determined as specified in Method D-1061-68 of ASTM "Book of Standards" (see 2.2).

4.4.1.2 Staining (wet and dry).- The test specimen for the stain test shall be two pieces of white, desized, cotton lawn cloth (80x80 print or finer), 4 by 4 inches, and four pieces of asbestos 4 by 4 inches. Place one 4 by 4 inch cotton lawn cloth between two specimens of asbestos. In turn lay this

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unit between two glass plates equal in size or larger than the test specimens. Then place this whole sandwich unit in a circulating air oven at a temperature of 300°F. ± 5°F. Carefully place a 20 pound weight in the middle of the top glass plate. The time of exposure shall be one hour to determine conformance with 3.2.2. For wet testing, the cotton lawn shall be completely saturated with water and then squeezed through a wringer to remove excess water before being laid between the two asbestos test specimens.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Padding and cover set.- One or a multiple of padding and cover sets of one description only shall be packaged in a snug-fitting fiberboard box conforming to style optional, type CF (variety SW), class domestic, grade 200 of PPP-B-636. Each box shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade optional of PPP-T-45.

5.1.1.2 Padding, metal mesh.- The metal mesh padding shall not require packaging.

5.1.2 Level C.- Padding and cover sets shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.-

5.2.1.1 Padding and cover sets.- Padding and cover sets of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, grade V2s of PPP-B-636; or overseas type, style A or I, grade A or B, type 2 load of PPP-B-601, as specified (see 6.2). Each fiberboard container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Each wood container shall be closed and reinforced in accordance with the appendix of the container specification; and shall be waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L-10547. The weight of the contents of each fiberboard container shall not exceed 65-pounds and for each wood container shall not exceed 150-pounds.

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5.2.1.2 Padding, metal mesh.- The metal mesh paddings shall be packed flat in a snug-fitting shipping container conforming to overseas type, style A or I, grade A or B, type 2 load of PPP-B-601. Multiple packs of paddings shall be separated physically from one another and shall be immobilized from movement while in transit by the application of water resistant material. The weight of contents, skids, closure, and reinforcing requirements shall be in accordance with the applicable requirements of PPP-B-601.

5.2.2 Level B.-

5.2.2.1 Padding and cover sets.- Padding and cover sets of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636; or domestic type, style A or I, grade B, type 2 load of PPP-B-601, as specified (see 6.2). Each fiberboard container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. Each wood container shall be closed and reinforced in accordance with the appendix of the container specification. The weight of the contents of each fiberboard shall not exceed 65-pounds and for each wood container shall not exceed 150-pounds.

5.2.2.2 Padding, metal mesh.- The metal mesh paddings shall be packed as specified in 5.2.1.2, except the shipping container shall conform to domestic type, style A or I, grade B, type 2 load of PPP-B-601. The material used for separating and immobilizing the contents need not be water resistant.

5.2.3 Level C.- Padding and cover sets, packaged as specified in 5.1, and metal mesh paddings shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages (when applicable) and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This padding and covering set is intended for covering 2, 4, 6 and 8 roll flatwork ironers.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type and class of padding required (see 1.2).
c. Make and type of flatwork ironer (see 3.5).

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d. Number and width of rolls (see 3.5).
e. Type and diameter of rolls (see 3.5)..
f. Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
g. Type of shipping container desired for level A or B packing of padding and cover sets (see 5.2.1.1 and 5.2.2.1).

6.3 This specification contains the requirements of Interim Purchase Description IP/DES S-14-0, dated 5 May 1970.

Custodians:

Army - GL
Navy - YD
Air Force - 84

User activities:

Army - MD
Navy - MC

Preparing activity:

Army - GL

Project No. 3510-0216

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

PADDINGS AND COVERS FOR FLATWORK IRONER ROLL MIL-P-43219A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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